Improving Compartmental Disease Models for Equation Learning

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IHBEM: Data-driven integration of behavior change interventions into epidemiological models using equation learning.

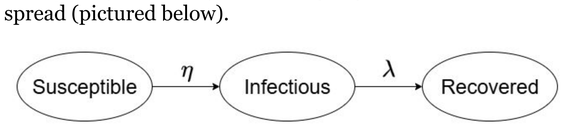
This research was supported in part by the National Science Foundation under grant DMS 2327836.

Introduction

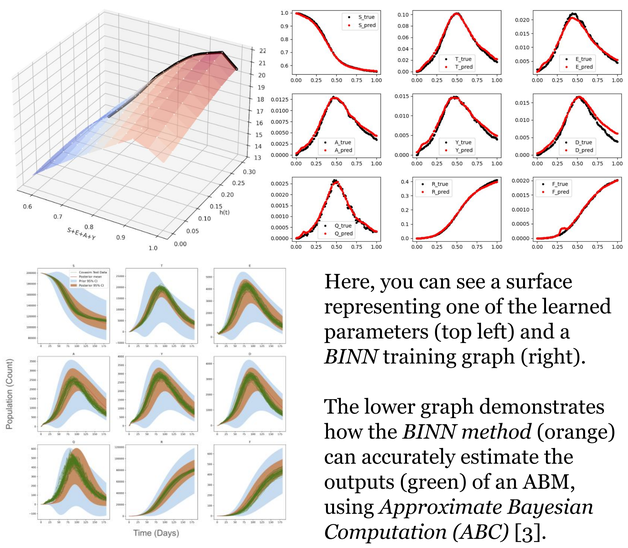
This project is part of an effort to create a new class of model which combines the strengths of two important model types:

- Agent-Based models
  - Track interactions between agents to identify trends
  - High fidelity
  - Computationally intensive
- Compartmental models
  - Track movement of a population through set of compartments, using differential equation approach
  - Lower Fidelity
  - Computationally efficient

A prominent example of a compartmental model is the susceptible, infectious, recovered (SIR) model for disease spread (pictured below).



We seek to combine the differential equation-based approach of the compartmental model with state-based parameters learned using the *Biologically-Influenced Neural Networks method (BINN)* [1]. This creates a new class of model with both the computational efficiency of the compartmental model and the fidelity of the ABM. [2]



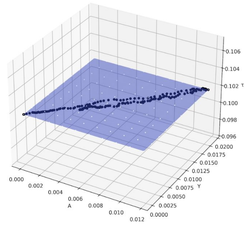
Here, you can see a surface representing one of the learned parameters (top left) and a *BINN* training graph (right).

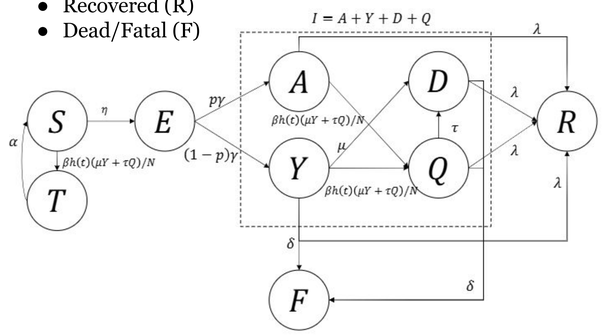
The lower graph demonstrates how the *BINN method* (orange) can accurately estimate the outputs (green) of an ABM, using *Approximate Bayesian Computation (ABC)* [3].

Model Architecture

Our previous work [2] involved using an improved SIR compartmental model to estimate the outputs of the open-source ABM *Covasim* [4], a COVID-19 disease spread simulation. The model includes both testing and quarantine interventions, to model their impacts. Here are the compartments, and the original model diagram obtained from [2].

- Susceptible (S)
- Susceptible and quarantined (T)
- Exposed, but not yet infectious (E)
- Infectious and asymptomatic (A)
- Infectious and symptomatic (Y)
- Diagnosed (D)
- Infectious and quarantined (Q)
- Recovered (R)
- Dead/Fatal (F)





Model Updates

Here, we contribute by changing the structure of the base compartmental model to address problems with the previous model. The updated model is pictured right, with the important changes labelled.

1 Previously, the Q and D compartments contained individuals with and without symptoms. This led to strange behavior, such as individuals being able to die without showing any symptoms. We split the D and Q compartments, replacing them with Q_A/D_A (asymptomatic) and Q_Y/D_Y (symptomatic).

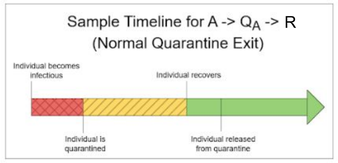
2 Previously, it was impossible for exposed individuals to be quarantined. We added a new Q_E compartment for this purpose, and allowed individuals to continue to the new Q_A and Q_Y compartments when they became infectious.

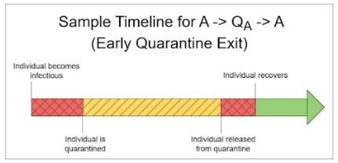
3 Modelling the τ parameter with the *BINN* method reveals that it is actually a constant (its surface is a flat plane, as seen left).

4 Asymptomatic individuals never got tested. We added a transition from A to D_A, with an appropriate testing rate determined by values from the Covasim settings.

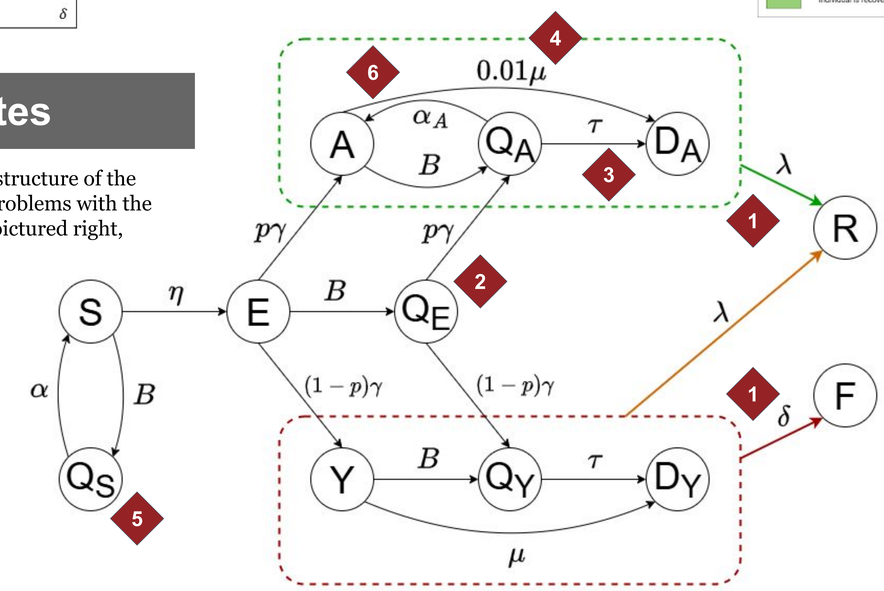
5 T was renamed to Q_S, to match Q_E, Q_A and Q_Y.

6 The model assumed all individuals recover/die before leaving quarantine. Asymptomatic individuals are often released early, with this being more common the shorter the quarantine period. We model this with a timeline (seen right).



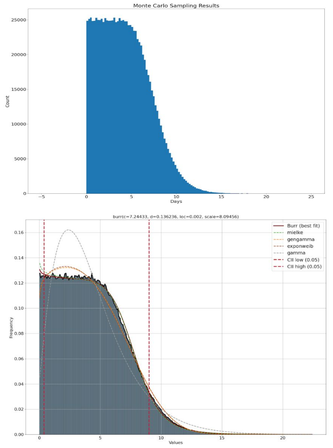






The time from quarantine to recovery (Q_A -> R) can be gleaned from distributions in *Covasim*. We use a Monte Carlo method to sample, and fit a distribution to the data. Here, a burr (type III) distribution was found to fit best.

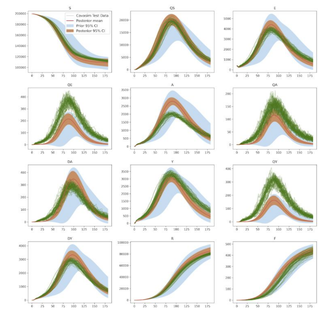
This distribution allows us to determine a rate α_A at which individuals will move from Q_A to A.



Updated Model Results

Changes 1-5 have been implemented to the model code (number 6 is WIP). Here is a current summary, with an *ABC* plot showing the performance:

- Model still estimates most compartments well
- Has a tendency to underestimate non-susceptible quarantine compartments
  - Could be due to a violation of the mean-field assumption the model uses
  - Needs further investigation
- Overestimates A compartments, compared with Y compartments
  - They likely have differing recovery rates that need to be accounted for



References

[1] Lagergren JH, Nardini JT, Baker RE, Simpson MJ, Flores KB. Biologically-informed neural networks guide mechanistic modeling from sparse experimental data. PLoS computational biology. 2020;16(12):e1008462.
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